Key Punch Data:	County Number	Source Number	Emission Point #		Action Type	Date Mo Day Yr	Agency/ Indiv. ID	
Line 1	0 1 0 (1–3)	0 2 3 0 (4–7)	0 0 0 0 3 (8–12)	0 0 0 (13–15)	9 9 7 (16–18)	2 (52)	0 4 / 2 8 4 (54–59)	9 0 9 (66–68)	N (80)

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1) 0 0 5 (16–18) 3 (78) C (80)

Quality as Received

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Company: Martin Marietta Cement

Inspected by: G. Mathews

Site Location: Lyons

County: Boulder

Contact Person: Roy Cook

Time: 2:00 - 4:15

Telephone No.: 534 - 4206

EIS Update: NA

Inspection: Routine [x] Complaint [ ] FA [ ]

1.75 Hrs.* Permit No. NA

Emission Points Subject to NSPS: NA

The facility has been purchased by Southwestern Portland Cement Co. I met w/ Mr. Doug Mac Iver to discuss offset calculations for final approval of C-12, 144, APEN requirements for change of ownership, and other items discussed in my Oct. 12, 1983 letter.

DOUGLAS Y. MAC IVER
MANAGER - ENVIRONMENTAL ENGINEERING

SOUTHWESTERN
PORTLAND CEMENT COMPANY
15000 SEVENTH STREET, SUITE 201
VICTORVILLE, CALIFORNIA 92392
(619) 245-1661
MAILING ADDRESS: P. O. BOX 937

APC-78 (Rev. 5-15-83)

* INCLUDE time for inspection, travel & write-up

AIR POLLUTION CONTROL - FIELD INSPECTION WORK SHEET

[illegible] County Number [illegible] Source Number [illegible] Emission Point [illegible] Action Type [illegible] Date Mo Day Yr [illegible] Agency/Indiv. ID

[illegible]

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1) 0 0 0 5 | 4 | C

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Quality as Received

Company: [illegible] | Inspected by: [illegible]

Site Location: [illegible] | County: [illegible]

Contact Person: [illegible] | Time: [illegible]

Telephone No.: | EIS Update: [illegible]

Inspection: Routine [ ] Complaint [ ] FA [ ] | 0.50 Hrs.* Permit No. [illegible]

Emission Points Subject to RGPS: All

[illegible] of emissions from finish product [illegible].
[illegible] on a
[illegible]
[illegible]
He [illegible] have it corrected immediately.
During conversation [illegible] that
[illegible] of the dryer
will [illegible].

By [illegible] emissions had ceased.

RECEIVED
NOV 19 1984
[illegible]
APCD

APC-78 (Rev. [illegible]) * INCLUDE time for inspection, travel & write-up

AIR POLLUTION CONTROL - FIELD INSPECTION WORK SHEET

Key Punch Data:		County Number	Source Number	Emission Point #		Action Type	Date Mo Day Yr	Agency/ Indiv. ID	
1	1	0 1 3	0 3 3 6	0 0 0 0 3	0 0 0	9 9 7	1 2	0 8 1 7 8 4	0 0 0	1
	2 3	4 7	8 12	13 15	16 18	52	54 59	66 68	8

001 - 012

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1) 0 0 5 (16 18) 1 (78) 0 (80)

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Quality as Received

Company: S. W. Portland Cement Company (Formerly Martin Marietta) Inspected by: G. Mathews

Site Location: P.O. Box 529 · Lyons, 80540 County: Boulder

Contact Person: Roy Cook, Sam Angelo Time: 8:30 - 11:45 am

Telephone No.: 534 - 4206 EIS Update: Attached

Inspection: Routine ☒ Complaint ☐ FA ☐ 8.25 Hrs.* Permit No.

Emission Points Subject to NSPS: Kiln (001) and clinker coolers (008)

Annual inspection per 1984 APCD Contract.

Annual production figures for 1983. See attached.

Cement produced w/ local mtls.: 413,500 tons.
Cement produced w/ Utah clinker: 33,952 Tons

Cement produced, total: 447,252 tons.

1. Point 001, Kiln Stack. (C-12,444 (2))

1983 fuels: Coal : 75,878 tons (0.4% Sulfur)
#2 Oil : 552,551 gal (0.41% Sulfur)
Gas : 271,230,000 SCF

Previously reported problem w/ baghouse has been corrected for at least 6 mo. by increasing reverse air flow.

Opacity: 0%

2. Point 002: Raw materials crushers

Primary crushers → baghouse : 0% opacity
Surge silo → baghouse : 0%
Conveyor → baghouse : 0%
Sec. crusher and waste dust silo → baghouse : 0%

OCT 1 8 1984

APC-78 (Rev. 5-15-83) * INCLUDE time for inspection, travel & write-up

S. W. Portland Cement, Source 3
-17-84 Inspection Report continued.

Point Ø02 continued

Fugitive emissions from enclosure around primary crusher : 10-15 % Opacity

At time of arrival, 8:30 am, a large cloud of dust was being generated in area of limestone dryer or waste dust silo. Roy Cole identified likely source as a clogged transfer point below the dryer. By 9:15 when we inspected area, no problems were noted.

3. Point Ø03, Limestone dryer.

1983 fuel: Gas burned : 178,732,000 SCF

0% Opacity.

SW Portland will be replacing Gortex filters with the reliable Nomex fiberglass bags. The Gortex gives out at higher temps.

SW Portland is planning to convert dryer to coal.

4. Point Ø04, A-frame and clinker storage.

A-frame doors: North has a curtain and south will be getting one; at time of inspection south door is completely open.

Outside storage: Still stockpiling clinker produced in Utah. When front loader drops clinker into hopper, >20% opacity emissions are produced.

5. Point Ø05, Site Fugitive Dust.

No problems noted. Vacuum vehicles and spray bars used. Haul roads are sprayed w/ water, not currently using chemical suppressant.

S.W. Portland Cement, Source 3
?-17-84 Inspection Report continued

6. Point 006, Coal handling.
No deliveries and crusher not operating at time of insp.

7. Point 007, Quarries.
Drilling rig shroud is in bad condition. This was noted last year. No drilling at time of inspection.
Observation of blast at this time, approx. 50 holes, 3'-15' deep, 3-4,000 lbs of explosives. Winds were from the south 10-15 mph. My observation point was near quarry and I was unable to view the off property line for the no off-prop. transport criterion. The cloud of dust went aloft rapidly and was visibly carried northward for some distance at increasing altitudes.

Point 008, Clinker coolers.
This is the only plant modification since the last inspections. The east and west clinker coolers have been ducted through a heat exchanger in order to protect the filters in the baghouses from excess heat (max. 350°F). Only one vent point is now set up for the 2 coolers. Opacity: 0%.
Handling of hot clinker is still a problem. Too many fines or a kiln flush can cause hot clinker to be dumped into a storage area with an open door. It's a large door for maneuvering a front loader. Both during clinker drop and load out excess (>20%) emissions are generated.

9. Point 009, Homogenizing silo
0% opacity

10. Point 010, Finished products silo
0% opacity

S. W. Portland Cement, Source 3
- 7 - 84 Inspection Report continued.

11. Point Φ11, Mill Building, 4 @ exhaust points.
2 @ east baghouse, raw mill 0% opacities
2 @ west baghouse, finish mill 0% opacities.

12. Point Φ12, Silica and iron ore siloes
Baghouse operates only during loading into siloes; not operating at time of inspection.

13. Facility has a number of outstanding permit items that must be addressed without further delay. Please refer to attached letter and previous letters to Martin-Marietta.

Status: Not in compliance.

10-3-84 [signature]

OCT 1 ...

S.W. Portland Cement

Emission Point ∅∅1 Type of Source Kiln Stack

Permit No. C-12,444 (2) Is a Revised APEN Required? yes change of ownership.

Remarks: 413,300 tons of cement produced in 1983

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMPL*
Part.	102 ton/yr.	C-12,444 (2)	413,300 tons/yr x 245 lb/ton x (1-0.999) baghouse = 50.63 tons/yr	SCC code 3-05-006-06 (p. 65)	YES
Opacity	20%	Reg. No. 1 II, A, 1	0%	Observation	YES

Emission Point ∅∅1 Type of Source Kiln Stack

Permit No. C-12,444 (2) Is a Revised APEN Required? yes. ownership change

Remarks: 75,878 tons of coal burned in 1983 (0.4% S); 552,551 gal. No. 2 fuel oil burned in 1983 (0.41% S)

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMPL*
SO_x	851 tons/yr.	C-12,444 (2)	Kiln feed 413,300 tons/yr x 10.2 lb/ton x (1-0.75) control factor x 1 ton/2000 lb = 527 ton/yr Coal combustion 75,878 tons/yr x 6.8 S lb/ton x 0.4% S x (1-0.75) x 1 ton/2000 lb = 26 ton/yr	3-05-006-06 AP42, 8.6-4 AP42, 8.6-4	

Continued

10-1-84 GM

APCD:106 (5-83) *If No, Indicate what action has been taken.

Emission Point ΦΦ1 Type of Source Kiln Stack

Permit No. ______ a Revised APEN Required? ______

Remarks: ______

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
S	continued		Fuel Oil combustion 552,551 gal/yr x 4.2 S lb/ton x 0.41 % S x (1 - 0.75) control factor x 1 ton/2000 lb = 119 ton/yr. Natural gas combustion 0 tons/yr Total : 671 ton/yr	AP42, 8.6-3 AP42, 8.6-4	YES

Emission Point ΦΦ1 Type of Source Kiln Stack

Permit No. ______ Is a Revised APEN Required? ______

Remarks: ______

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
NO_x	None specified	None	413,300 tons/yr x 2.6 lb/ton x 1 ton/2000 lb = 537 ton/yr	3-05-006-06	
VOC	None specified	None	0	3-05-006-06	
CO	None specified	None	0		

APCD:106 (5-83) *If No, indicate what action has been taken.

Emission Point ØØ2 Type of Source Raw materials crushers
Permit No. Grandfathered Is a Revised APEN Required? YES ownership change
Remarks: 747,577 tons processed in 1983

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	20% opacity	Reg No. 1, III, D, 2, c	10-15% out of primary crusher enclosure	Observation	YES
Part.	Design Rate 200 TPH PE = 17.3 (200)$^{0.16}$ = 40.38 lb/hr PRIMARY CRUSHER	Reg. No. 1 III, C, 1, b	747,577 tons/yr x 0.5 lb/ton x (1 - 0.999) x (1/2000) = 0.19 tons/yr x (1/7896 hrs) = 0.05 lb/hr.	3-05-006-09	YES
	RAW MTL UNLOADING	None	747,577 tons/yr x 0.2 lb/ton x (1/2000) = 74.8 tons/yr	3-05-006-07	

Emission Point ____ Type of Source ____
Permit No. ____ Is a Revised APEN Required? ____
Remarks: ____

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
	RAW MTL PILES	NONE	Unknown quantity	3-05-006-08	
	RAW MTL TRANSFER	NONE	747,577 tons/yr x 0.3 lb/ton x (1 - 0.99) totally enclosed w/ neg. pressure to baghouse x (1/2000) = 1.12 tons/yr.	3-05-006-02 and APCD FD Memo	

*If No, indicate what action has been taken.

Emission Point ϕ03 Type of Source Limestone dryer

Permit No. C-12,444 (1) a Revised APEN Required? 0

Remarks: 178,732,000 SCF gas burned in 1983

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	144 ton/yr.	C-12,444 (1).	178.7 x10^6 ft^3 gas/yr x 3.0 lb/10^6 ft^3 x (1 - 0.999) baghouse x (1/2000) = 3x10^-2 ton/yr	1-02-006-02	
SO_x			178.7 x10^6 ft^3/yr x 0.6 lb/ft^3x10^6 x (1/2000) = 0.05 ton/yr.		
NO_x			178.7 x10^6 ft^3/yr. x 140 lb/10^6 ft^3 x (1/2000) = 5.5 ton/yr.		

Emission Point ______ Type of Source ______

Permit No. ______ Is a Revised APEN Required?* ______

Remarks: ______

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
VOC			178.7 x10^6 ft^3/yr x 2.8 lb/ton x (1/2000) = 0.25 ton/yr		
CO			178.7 x10^6 ft^3/yr. x 35 lb/ton x (1/2000) = 3.1 ton/yr.		

OCT

APCD:106 (5-83)

*If No, Indicate what action has been taken.

S.W. Portland

Emission Point ØØ3 Type of Source Limestone dryer

Permit No. C-12,444 (1) Is a Revised APEN Required? Yes - ownership

Remarks: 413,300 tons of cement produced in 1983

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMPL.*
Part.	144 ton/yr	C-12,444(1)	413,300 ton/yr × 64 lb/ton × (1-0.999) baghouse × (1/2000) = 13.23 ton/yr	3.05.006.13	YES

Emission Point ØØ4 Type of Source A-frame and clinker storage

Permit No. None Is a Revised APEN Required?

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMPL.*
Part.	20%	1, III, A, 1	>> 20% at outside storage hopper	Observation	NO

APCD:106 (5-83) *If No, indicate what action has been taken.

S. W. Portland

Emission Point ØØ5 Type of Source Site Fugitive dust

Permit No. None Is a Revised APEN Required? No

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	No off-property transport guideline	1, III, D, 2, e	No off-property transport Vehicles 90 miles/day x 250 days/yr. x 3.0 lb/mi x (1.0 - 0.50) x (1/2000) = 16.9 tons/yr	Observation 1982 data 3-05-020-11 watering (no credit)	YES

Emission Point ØØ6 Type of Source Coal handling

Permit No. Grandfathered Is a Revised APEN Required?

Remarks: 75,878 tons of coal burned in 1983

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	Unknown.		No emission factor has been identified for coal crushing/handling.		

APCD:106 (5-83)

*If No, indicate what action has been taken.

Type of Source [illegible]

Permit No. Grandfathered Is a Revised APEN Required? ~~NO~~ YES - FD. Control Plan

Remarks: 703,034 tons of [illegible] rock

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
P	20% and No off-property transport	Reg. No. 1, III, D, 2, d.	variable 0-20% Unable to observe property line transport.	Observation	YES
			BLASTING 703,034 tons/yr. × 0.16 lb/ton × (1/2000) =56.2 tons/yr.	3-05-020-09 (p. 28)	

Emission Point ØØ8 Type of Source Clinker cooler baghouse

Permit No. Grandfathered Is a Revised APEN Required? YES - ownership

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	0.10 lb/ton feed to kiln	Reg. No. 6, IV, C, 2, a.	413,300 ton/yr × 1.0 lb/ton × (1 - 0.999) baghouse × (1/2000) = 0.21 ton/yr. ≤ 0.10 lb/ton feed	EIS printout	YES.
	10% opacity	Reg. No. 6, IV C 2 b	0%	Observation	YES.

APCD:106 (5-83)

*If No, indicate what action has been taken.

S. W. Portland [illegible] [illegible]

Emission Point ØØ9 Type of Source Homogenizing silo

Permit No. Grandfathered Is a Revised APEN Required? y - ownership

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	20%	Reg. No. 1, II. A. 1.	0% No emission factor listed in Coke Book.	Observation	YES

Emission Point Ø1Ø Type of Source Finished products silo.

Permit No. Grandfathered. Is a Revised APEN Required? YES. Ownership.

Remarks: 447,252 tons of cement processed in 1983

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	20%	1, II, A. 1.	0%	Observation	yes.
			447,252 tons/yr x 0.24 lb/ton x (1 - 0.999) baghouse x (1/2000) = 0.05 ton/yr.	3-05-006-19	

APCD:106 (5-83)

*If No, Indicate what action has been taken.

Emission Point ∅011 Type of Source Mill Building

Permit No. Grandfathered Is a Revised APEN Required? No - Observation

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	20%	1, II, A, 1	0%	Observation	YES.
			417,252 ton/yr x 32.0 lb/ton x (1- 0.999) baghouse x (1/2000) = 7.2 tons/yr.	3.05.006.17	

Emission Point ∅12 Type of Source Silica silos

Permit No. Grandfathered Is a Revised APEN Required?

Remarks:

POLLUTANT	ALLOWABLE	REFERENCE	ACTUAL	REFERENCE	IN COMP?*
Part.	20% Opacity	1, II, A, 1	Not observed No emission factor listed in SCC Code book		Unknown

APCD:106 (5-83) *If No, indicate what action has been taken.

OPERATIONS REPORT - LYONS PLA[illegible]

CORRECTED COPY

	MONTHLY			YEAR-TO-DATE		
Rental Drill	Ft: 0	Hrs: 0	Ft/Hr: 0	Ft: 11,15[illegible]	Hrs: 201.5	Ft/Hr: 55.4
I.R. Drill	Ft: 5,676	Hrs: 61.5	Ft/Hr: 92.3	Ft: 105,514	Hrs: 1300.5	Ft/Hr: 81.1
Stripping	Tons: 66,424			Tons: 1,119,313		

CRUSHING:	TONS	HOURS (744)	TONS/HR.	% OPER.	TONS	HOURS (8760)	TONS/HR.	% OPER.
Hi.Cal.R#1	23,770	199.1	119.4	26.8	320,018	2418.0	132.3	27.6
Hi.Cem.R#3	19,404	162.5	119.4	21.8	227,804	1724.6	132.1	19.7
Lo Cem.R#3-4	3,382	28.3	119.4	3.8	98,938	745.6	132.7	8.5
Weathered	6,988	58.4	119.6	7.8	56,274	425.4	132.3	4.9
R. #4	0				0			
Silica	1,482	11.4	130.0	1.5	11,164	84.8	131.6	1.0
Gypsum	1,597	12.3	130.0	1.7	23,657	179.4	131.9	2.0
Iron *est[illegible]*	1,042	8.0	130.0	1.1	8,722	74.1	131.2	.8
TOTALS	57,665	480.0	120.1	64.5	747,577	5651.9	132.3	64.5

	TONS	HOURS	TONS/HR.	% OPER.	TONS	HOURS	TONS/HR.	% OPER.
DRYER	57,665	480.0	120.1	64.5	747,577	5651.9	132.3	64.5
RAW GRIND	56,068	533.0	105.2	71.6	714,312	6223.2	114.8	71.0
KILN	35,532	558.9	63.6	75.1	413,300	7131.2	58.0	81.4
FINISH MILL:								
I	6,035	64.9	92.9	8.7	175,454	1917.2	91.5	21.9
I LA	0				0			
II	12,862	138.4	92.9	18.6	181,325	1989.3	91.2	22.7
II LA	-48				3,980	43.3	91.9	.5
II MOD	4,368	46.6	93.7	6.3	55,302	607.8	90.9	6.9
Class G	0				0			
PR[illegible]ICT	1,850	38.7	47.8	5.2	25,861	521.0	49.5	5.9
M[illegible]	353	5.7	61.5	.7	5,390	85.2	63.3	1.0
[illegible]ALS	25,420	294.3	86.4	39.5	447,252	5163.8	86.6	58.9

Equiv. Prod: .85x 35,532 +.15x 25,420 =34,015 Tons | .85x 413,300 +.15x 447,252 =418,393 Tons

186.27 | *178.26*

POWER:	KWH: 6,336,000	COST/KWH: 0.0403	KWH: ~~?4,?3?,000~~ *74,933,000*	COST/KWH: ~~0.0411~~ *0.029*
	COST: 255,598.09	COST/EQ.TON: 7.51	COST: 2,230,957.91	COST/EQ.TON: 5.33
			(Avg.989)	(Avg.10,633)

FUEL: (974)	GAS MCF	OIL/GAL	COAL/TONS	BTU/TON	GAS MCF	OIL/GAL	COAL/TONS	BTU/TON
DRYER	16,643	-	-	281,111	178,732	-	-	236,452
KILN (Coal 10,300)	24,720	90,597	5,635	4,324,475	92,498	552,551	75,878	4,324,781
TOTALS	41,363	90,597	5,635	4,780,691	271,230	552,551	75,878	4,752,475

MANHOURS:	NUMBER	HOURS	MANHOURS/EQ.TONS	NUMBER	HOURS	MANHOURS/EQ.TONS
Hourly	86	16,034	.47		200,502	.48
Staff	41	7,325	.22		86,489 ✓	.21
TOTALS	127	23,359	.69		286,991 ✓	.69

SALES:	MONTHLY	YEAR-TO-DATE
I	5,849	176,522
I LA	0	0
II	7,766	188,224
II LA	156	3,139
"G"	0	0
II MOD	2,216	53,113
P[illegible]CT	1,571	26,054
M[illegible]	196	5,308
Other	0	173
TOTALS	17,754	452,533

SAFETY:	MONTHLY	YEAR-TO-DATE
First Aid Cases	14	157
Doctor Cases	1	30
Lost Time Accidents	1+1adj.	9
Days since last lost time accident	1	1

Gerald T Anderson

Manager of Operations

MONTH: DECEMBER 1983

CORRECTED COPY

L-272 (7/83)